

bc240221856**Ayesha Mushtaq****Question:1**

Prove that $A - B = A \cap B^c$ by using Membership Table.

Solution:

A	B	A-B	B^c	$A \cap B^c$
1	1	0	0	0
1	0	1	1	1
0	1	0	0	0
0	0	0	1	0

Hence, proves that **L.H.S=R.H.S**

Question: 2

Using a truth table, determine whether the following statement is a tautology or a contradiction ?

$$(p \wedge q) \vee (\sim p \vee (p \wedge \sim q))$$

Solution:

p	q	$\sim p$	$\sim q$	$p \wedge \sim q$	$\sim p \vee (p \wedge \sim q)$	$p \wedge q$	$(p \wedge q) \vee (\sim p \vee (p \wedge \sim q))$
T	T	F	F	F	F	T	T
T	F	F	T	T	T	F	T
F	T	T	F	F	T	F	T
F	F	T	T	F	T	F	T

The given statement is **Tautology**.

Question: 3

Construct a truth table of the following statement.

$$(p \vee q) \vee (\sim p \wedge q) \rightarrow q$$

Solution:

The truth table is given below:

p	q	$p \vee q$	$\sim p$	$\sim p \wedge q$	$(p \vee q) \vee (\sim p \wedge q)$	$(p \vee q) \vee (\sim p \wedge q) \rightarrow q$
T	T	T	F	F	F	T
T	F	T	F	F	F	T

F	T	T	T	T	F	T
F	F	F	T	F	F	T

Question: 4

Using truth table, determine the validity of the argument.

$$p \wedge q \rightarrow \sim r$$

$$p \vee \sim q \rightarrow \sim q$$

$$\rightarrow p$$

$$\therefore \sim r$$

Solution:

p	q	r	$\sim q$	$\sim r$	$p \wedge q$	$p \wedge q \rightarrow \sim r$	$p \vee \sim q$	$\sim q \rightarrow p$	$\sim r$
T	T	T	F	F	T	F	T	T	F
T	T	F	F	T	T	T	T	T	T
T	F	T	T	F	F	T	T	T	F
T	F	F	T	T	F	T	T	T	T
F	T	T	F	F	F	T	F	T	F
F	T	F	F	T	F	T	F	T	T
F	F	T	T	F	F	T	T	F	F
F	F	F	T	T	F	T	T	F	T

Arguments of **row 2** and **row 4** are valid because the conclusion is true when all the premises are true.
